Project Report: Wildlife Sanctuary Navigator

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Domain: Artificial Intelligence (Informed Search)

Tools: Python, NetworkX , Matplotlib, A* Algorithm

1. Abstract

The Wildlife Sanctuary Navigator is a graph-based pathfinding system designed to optimize navigation within a wildlife sanctuary. Unlike traditional navigation systems, this model focuses on terrain awareness and environmental constraints, such as steep paths and restricted wildlife zones.

By implementing the A* Search Algorithm with a Euclidean heuristic, the system calculates the most efficient and practical route between two locations. The project demonstrates how AI can be applied to real-world scenarios involving environmental navigation and decision-making.

2. Problem Statement & Motivation

Navigating large wildlife sanctuaries presents several challenges:

Complex Terrain: Sanctuaries consist of uneven paths, forests, and elevated regions.

Wildlife Restrictions: Certain areas may be restricted due to animal movement (e.g., tiger zones).

Lack of Smart Routing: Traditional maps do not consider terrain difficulty or safety.

Efficiency vs Safety: Users need routes that are not just short, but also practical and safe.

This project aims to create an intelligent navigation system that accounts for both distance and terrain conditions.

3. System Architecture & AI Methodology

3.1 Graph Representation

The sanctuary is modeled as an Undirected Weighted Graph $G = (V, E)$:

Nodes (V): Represent key locations such as Park Entrance, Camps, and Wildlife Zones

Edges (E): Represent paths between locations

Weights (W): Calculated based on:

Distance × Terrain Factor

3.2 The A* Search Algorithm

The system uses the A* algorithm defined by the evaluation function:

$$f(n) = g(n) + h(n)$$

g(n) (Path Cost): Actual cost from start node to current node

h(n) (Heuristic): Estimated cost to reach goal

Heuristic used:

$h(n) = \sqrt{[(x2 - x1)^2 + (y2 - y1)^2]}$

This is Euclidean distance, ensuring the heuristic is admissible and the solution is optimal.

3.3 Innovation: Terrain-Based Cost Factor

A key feature of this project is the inclusion of terrain-based cost adjustment:

Flat Paths: Factor = 1.0

Steep Paths: Factor = 1.5

This allows the system to:

Avoid difficult terrain for general users

Reflect real-world navigation complexity

4. Implementation Details

The project is implemented in Python using:

NetworkX: For graph creation and A* pathfinding

Matplotlib: For visualization of the sanctuary map

Math Module: For heuristic (distance) calculations

The system takes user input for start and end locations and computes the optimal path accordingly.

5. Results & Discussion

5.1 Efficiency Comparison

The A* algorithm was compared with other search techniques:

BFS (Breadth-First Search):

Finds path with minimum steps

Ignores weights → inefficient in real scenarios

Dijkstra Algorithm:

Finds optimal path

Explores more nodes → slower

A* Algorithm:

Uses heuristic to guide search

Explores fewer nodes

Provides faster and optimal results

5.2 Path Analysis Example

For navigation between Park Entrance and Tiger Zone, the algorithm evaluates multiple paths:

If a shorter route includes steep paths (higher cost),

The system may choose a longer but flatter route

This demonstrates how the model prioritizes practical efficiency over pure distance.

6. Conclusion & Future Scope

The Wildlife Sanctuary Navigator successfully demonstrates the application of informed search algorithms in real-world navigation problems. It highlights that the shortest path is not always the most efficient when terrain and environmental factors are considered.

Future Enhancements

Real-time wildlife tracking integration

Risk-aware routing (avoiding dangerous zones)

GPS-based real-world implementation

Web or mobile application deployment

Multi-destination route optimization

7. Final Summary

This project bridges theoretical AI concepts with practical implementation. By modeling a wildlife sanctuary as a state-space graph and applying a cost-sensitive A* algorithm, the system provides intelligent route planning based on terrain and environmental conditions.

It demonstrates how AI can enhance navigation systems beyond traditional shortest-path approaches.

Final Deliverables

Source Code: Python implementation using NetworkX and Matplotlib

Digital Map: Coordinate-based graph representing sanctuary locations

Heuristic Model: Euclidean distance ensuring optimal pathfinding